

GATE

ALL BRANCHES



General Aptitude

ANALYTICAL APTITUDE

Lecture No.- 01



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Recap of Previous Lecture



Topic

Interest



Topics to be Covered



Topic-1

Blood Relations



[MCQ]



#Q. What is the compound interest on ₹2500 for two years at the rate of interest 4% per annum?

Assignment

A ₹ 180

B ₹ 210

C ₹ 204

D ₹ 220

$$\begin{array}{r} 2500 \times 1.04 \times 1.04 \\ 2704 \\ - 2500 \\ \hline 204 \end{array}$$

$$C.I. = P \left[\left(1 + \frac{r}{100} \right)^n - 1 \right]$$
$$C.I. = 2500 \left[\left(1 + \frac{4}{100} \right)^2 - 1 \right]$$
$$= 2500 \left[\left(\frac{1.04}{1} \right)^2 - 1 \right]$$
$$= 2500 \times 0.0816 = 204$$

[MCQ]



#Q. A sum of money was invested at simple interest at a certain rate for 3 years.
Had it been invested at a 4% higher rate, it would have fetched ₹480 more.
Find the principal.

Assignment

A ₹ 3000

B ₹ 4000

C ₹ 5000

D ₹ 6000

4% — 160

100% — ?

$$160 \times \frac{100}{4}$$

$$= 4000$$

₹160/year

[MCQ]



#Q. In what time a sum of money will double itself at a rate of S.I. of 6% per annum?

Assignment

$$A = P \left(1 + \frac{TR}{100} \right)$$

$$2P = P \left(1 + \frac{T \times 6}{100} \right)$$

$$1 = \frac{T \times 6}{100}$$

$$\therefore \underline{T} = \frac{100}{6} = \frac{50}{3} = 16 \frac{2}{3} \text{ yrs}$$

A 6 $\frac{1}{3}$ years

B 16 $\frac{2}{3}$ years

C 33 $\frac{1}{3}$ years

D 21 years

[MCQ]



#Q. In how many years will a sum of ₹800 at 10% per annum compounded semi-annually become ₹926.10? ^P_✓

Assignment

$$n = n$$

$$R = 5\% \text{ per semi-annually}$$

$$A = P \left(1 + \frac{R}{100}\right)^n$$

$$\Rightarrow \left(\frac{21}{20}\right)^3 = \left(\frac{21}{20}\right)^n$$

$$\therefore n = 3$$

$$\left(1\frac{1}{2}\right) \text{ yrs}$$

$$\Rightarrow 926.10 = 800 \left(1 + \frac{5}{100}\right)^n$$

$$\Rightarrow \frac{926.1}{800} = \left(\frac{105}{100}\right)^n$$

$$\Rightarrow \frac{9261}{8000} = \left(\frac{21}{20}\right)^n$$

A 1 $\frac{1}{3}$ years

B 1 $\frac{1}{2}$ years

C 2 $\frac{1}{3}$ years

D 2 $\frac{1}{2}$ years

[MCQ]

$$R = 4\%$$

$$D_{\text{diff}} = P \left(\frac{r}{100} \right)^2$$

$$A = P \left(\frac{r}{100} \right)^n$$
$$= P \left(\frac{104}{100} \right)^4$$



#Q. The difference between simple interest and compound interest on a sum for 2 years at 8% when the interest is compounded annually is ₹16. If the interest were compounded half yearly the difference in two interests would be nearly

C.I.

$$16 = P \left(\frac{8}{100} \right)^2$$

S.I.

Assignment

A

₹ 21.25

B

₹ 24.64

C

₹ 27.85

D

₹ 29.94

$$A = 2500 \times 1.04 \times 1.04$$
$$\times 1.04 \times 1.04$$

$$16 = P \times \frac{64}{10000}$$

$$= 2924.64$$
$$P = \frac{10,000}{4} = 2500$$

$$I = 424.64$$

$$I = \frac{2500 \times 4 \times 4}{100}$$

$$I = 25 \times 16 = 400$$

$$D_{\text{diff}} = 424.64 - 400$$
$$= 24.64$$

[MCQ]



#Q. Raju invested Rs.5488 for 3 years at C.I. and received an amount of Rs.6750 on maturity. What is rate percent?

A $7\frac{1}{2}\%$

B $9\frac{2}{3}\%$

C $3\frac{1}{3}\%$

D $6\frac{1}{3}\%$

$$A = P \left(1 + \frac{r}{100} \right)^n$$

$$\Rightarrow 6750 = 5488 \left(1 + \frac{r}{100} \right)^3$$

$$\Rightarrow \frac{6750}{5488} = \left(\frac{100+r}{100} \right)^3$$

$$\Rightarrow \left(\frac{15}{14} \right)^3 = \left(\frac{100+r}{100} \right)^3$$

Assignment

$$\Rightarrow \frac{15}{14} = \frac{100+r}{100}$$

$$\Rightarrow 1500 = 1400 + 14r$$

$$14r = 100$$

$$r = \frac{100}{14}$$

$$= 7\frac{1}{7}\%$$

Blood Relations



Maternal
Paternal

Great G.F. / Great G.M.
Mother / Father
forefather
Brother / Sister
G.F. / G.M.
Uncle / Aunt

Son / Daughter
Nephew / Niece
Cousins
husband / wife
Grand son / Grand daughter



Blood Relations

Maternal ✓
Paternal ✓
A is son of B.

Mother : Father

Brother : Sister

Uncle : Aunt

Grand Father : Grand Mother

Grand son : Grand daughter

(parent)

(sibling)

(Grandparent)

(Grand child)

Son : Daughter

Cousins

Nephew : Niece

Husband : Wife

(child)

(Spouse)

✓ Great Grand Father : Great Grand Mother
Forefather / Ancestors

Generations:

$\boxed{G.F | G.T} \Rightarrow \boxed{G.G.P | G.G.T}$

→ Generation

(+)
(-)

$\boxed{Mother | Father} \Rightarrow \text{Uncle/Aunt}$

husband ^{wife}
A ~ B

Cousins $\leftarrow \boxed{Brother | sister} \rightarrow$ Spouse

A $\leftrightarrow$ B

Like Men are
always
right

$\boxed{Son | daughter} \Rightarrow \text{Nephew/Niece}$

Grand son / Grand daughter



In laws:

Depends

Son-in-law
daughter-in-law

father-in-law
mother-in-law
brother-in-law
sister-in-law

~~By Birth~~



Before



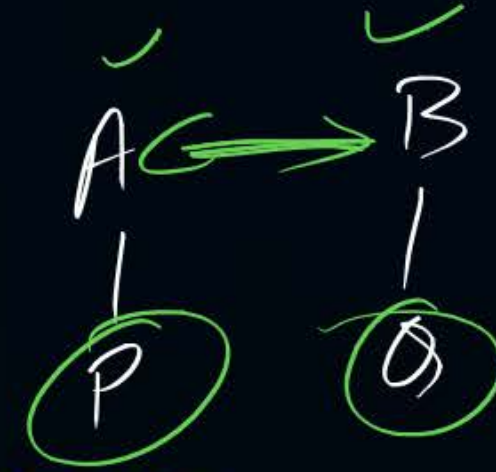
After

In laws:



✓
Spouse Mother = Mother in law
Spouse Father = Father in law
Spouse Brother = Brother in Law
Spouse Sister = Sister in law

~~Spouse son = son~~
~~Spouse daughter = daughter~~



Son in Law
Daughter in Law

[MCQ]



#Q. Pointing to a boy in the photograph Manasa said, "His sister is the only daughter of my father". How is the boy related to Manasa's father?

me



Son

A

Father

B

Brother

C

Son

D

Cousin

[MCQ]



#Q. Pointing to Neha, Mahesh said, "Her mother's only daughter is my daughter". How is Mahesh related to Neha?

She

Father

- A** Brother
- B** Son
- C** Uncle
- D** Father

[MCQ]



#Q. A is brother of B. B is wife of C.

C is son of D. D is wife of F.

Then what is B to F?

Assignment

- A** Daughter-in-law
- B** Cousin
- C** Son-in-law
- D** None of these

[MCQ]



#Q. Pointing towards a person in a photograph, Raman said, “She is the only daughter of the mother of my brother’s sister”. How is that person related to Raman?

Assignment

- A** Father
- B** Mother
- C** Uncle
- D** Sister

[MCQ]



#Q. If Neena says, “Anita’s father Raman is the only son of my father-in-law Mahipal”, then how is Bindu, who is the sister of Anita, related to Mahipal?

Assignment

- A** Daughter
- B** Wife
- C** Daughter-in-law
- D** None of these

[MCQ]



#Q. Pointing to a photograph, a woman says, "This man's son's sister is my mother-in-law." How is the woman's husband related to the man in the photograph?

Assignment

A

Son

B

Grand son

C

Son-in-law

D

Nephew

[MCQ]



#Q. When Amit saw Manoj, he recalled and said “He is the son of the father of the mother of my daughter”. What is Manoj to Amit?

Assignment

A

Cousin

B

Brother-in-law

C

Uncle

D

Brother

[MCQ]

#Q. Direction:

$X+Y$ means X is the father of Y,

$X-Y$ means X is the wife of Y,

$X \times Y$ means X is the brother of Y,

$X \div Y$ means X is the daughter of Y.

Assignment

$A \times B - C$ means _____

A

A is husband of C

B

A is brother-in-law of C

C

A is son of C

D

Can't be determined

[MCQ]

#Q. Direction:

$X+Y$ means X is the father of Y,

$X-Y$ means X is the wife of Y,

$X \times Y$ means X is the brother of Y,

$X \div Y$ means X is the daughter of Y.

Assignment

In $A \times B \div C$, which of the following statement is true?



A is father of C



A is brother of C



A is uncle of C



A is son of C

[MCQ]



#Q. Rohit said to Mohit, "The boy playing basket ball is younger brother of the two brothers of the daughter of my father's wife". How is the boy playing basket ball related to Rohit?

Assignment

- A** Brother-in-law
- B** Nephew
- C** Cousin
- D** Brother

[MCQ]



#Q. Introducing Swapna, Madhavi said, "She is the only daughter of my father's only daughter". How is Madhavi related to Swapna?

Assignment

- A** Niece
- B** Aunt
- C** Cousin
- D** None of these

[MCQ]



#Q. A, the son of B was married to C, whose sister D was married to E, the brother of A. How is D related to B?

Assignment

- A** Sister
- B** Cousin
- C** Sister-in-law
- D** Daughter-in-law



2 mins Summary



Topic

Blood Relation



THANK - YOU